

2ND MIDTERM EXAMINATION
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
 II Semester, April/May, 2026

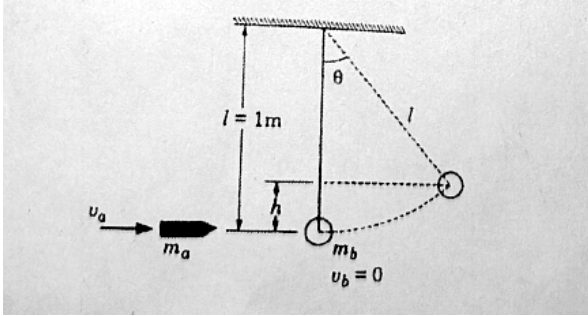
Paper Code: ES 114

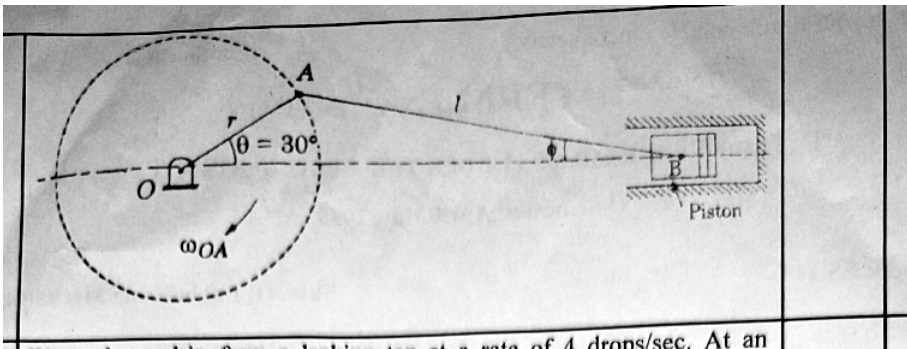
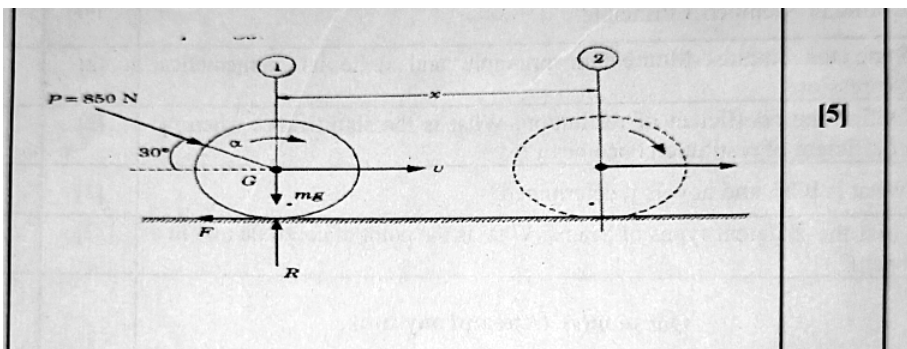
Subject: Engineering Mechanics

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory, and any two more questions from the remaining.

Q. No.	Question(s)	Max. Marks	CO(s)
1(a)	Define D'Alembert's Principle.	[2]	3
1(b)	State the Impulse-Momentum principle and write its mathematical expression.	[2]	3
1(c)	Define the coefficient of restitution. What is the significance when the coefficient of restitution becomes 1?	[2]	3
1(d)	What is ICR, and how is it determined?	[2]	4
1(e)	List the different types of beams. What is the point of contraflexure in a beam?	[2]	4
Question(s) (Attempt any two)			
2	Bullet A of mass 0.01 kg moving with a velocity of 100 m/s hits a bob B of a simple pendulum horizontally as shown in the Figure. Find the maximum angle through which the pendulum swings when: (a) The bullet gets embedded in the bob (b) The bullet rebounds from the surface of the bob with a velocity of 20 m/s (c) The bullet escapes from the other end of the bob with a velocity of 20 m/s 	[10]	3

Q. No.	Question(s)	Max. Marks	CO(s)
3	A reciprocating engine mechanism is shown in Figure below. The crank OA has a constant angular velocity of N = 300 r.p.m. The crank OA is of length r = 12.0 cm and the connecting rod AB is of length l = 60 cm. Find: (a) the angular velocity of the connecting rod and (b) the velocity of the piston when the crank makes an angle of $\theta = 30^\circ$ with the horizontal. 	[10]	4
4(a)	Water drops drip from a leaking tap at a rate of 4 drops/sec. At an instant, what is the vertical separation between two consecutive drops if the lower of the two drops has a velocity of 50 m/s?	[5]	3
4(b)	A roller of mass m = 600 kg and radius r = 0.25 m is pushed with a constant force P = 850 N on a rough horizontal plane as shown in Fig. If the roller starts from rest and rolls without slipping, find the distance it must roll to acquire a velocity of 3 m/s. 	[5]	4

**** End of Question Paper ****